Technology in Society 82 (2025) 102909

Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc

The role of explainable artificial intelligence (XAI) in innovation processes: a knowledge management perspective

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ARTICLE INFO

Keywords:
Explainable artificial intelligence
Innovation management processes
Knowledge management mechanisms
Human-machine collaboration
Creativity and innovation techniques
Creative process

ABSTRACT

This study investigates the role of Explainable Artificial Intelligence (XAI) in supporting knowledge management during the innovation process. While traditional AI is recognized for sustaining innovation via enhancement of different knowledge management processes, little is known about how XAI specifically impacts these processes by making AI outputs more interpretable for humans. To address this gap, we conduct a single case study on Intellico, an XAI-driven organization specializing in knowledge management solutions for innovation. The study's primary contribution is a model on how XAI facilitates knowledge management along the four creative phases of innovation, namely preparation, incubation, illumination, and verification. The findings challenge the traditional alignment between specific knowledge processes and innovation phases, offering new theoretical insights into the interplay between XAI, knowledge management, and innovation. Additionally, the study provides practical guidance for organizations willing to leverage human-XAI collaboration for innovation purposes.

1. Introduction

Explainable artificial intelligence (XAI) is a branch of artificial intelligence (AI), distinct from traditional AI, focused on developing methodologies, tools, and frameworks that make AI systems transparent, interpretable, and comprehensible to humans (Tchuente et al., 2024). Unlike traditional AI, which often functions as a *"black box"* with little visibility into how decisions or predictions are made, XAI emphasizes clarity in the decision-making process (Meske et al., 2022). Hence, the key characteristics that distinguish XAI from conventional AI include: transparency, which reveals the internal logic or structure of the model (Sadeghi et al., 2024); interpretability, which translates complex outcomes into human-understandable terms (Kim et al., 2023); and justifiability, which provides evidence and rationale for specific decisions (Minh et al., 2022). These attributes make XAI particularly valuable for the promotion of trust in AI results, that is fundamental for effective human-AI collaboration. There are two main reasons for this. First, through clear and rational explanations, XAI enhances users' ability to critically evaluate and integrate AI outputs with their own expertise. As a result, users are reassured about XAI's decisions and human-machine collaboration is strengthened (Kolbjørnsrud, 2024). Second, by making models interpretable, XAI facilitates more accurate diagnostics of biases, errors, and potential unintended behaviors embedded in training data or algorithmic logics. This capability ensures more reliable and efficient AI systems (Meske et al., 2022).

Through its intrinsic nature, XAI builds confidence in the system and ensures that its knowledge is trustworthy for humans. This trust is particularly significant in the context of innovation, where the traditional knowledge sharing among humans for the generation of innovative outputs can be augmented through a new practice, i.e., collaborative knowledge sharing between humans and machines. Indeed, it is recognized that XAI is able of opening unprecedented opportunities for innovation, by offering new ways for organizations to capture, structure, and utilize knowledge effectively (Tchuente et al., 2024). Specifically, traditional black box AI systems often require extensive technical knowledge to operate. This limits their utility to a narrow set of AI specialists and complicates the incorporation of feedback from innovation experts, who may be non-AI experts (Meske et al., 2022). Conversely, XAI democratizes the access to advanced AI-generated knowledge, empowering non-AI experts to merge their

This article is part of a special issue entitled: Role of Knowledge Management published in Technology in Society.

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https://doi.org/10.1016/j.techsoc.2025.102909
Received 29 January 2025; Received in revised form 20 March 2025; Accepted 13 April 2025
Available online 14 April 2025
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expertise and creativity with sophisticated and interpretable AI models (Tchuente et al., 2024). In this way, a virtuous cycle of continuous feedback between humans and machines is promoted, thus enabling an innovation process based on a real synergy between humans and AI (Kolbjørnsrud, 2024). Specifically, innovation emerges from the collaboration between humans and XAI because this synergy combines the strengths of both parties in unique and complementary ways. Humans bring domain expertise, creativity, and contextual understanding, while XAI contributes clear data-driven insights, transparent pattern recognition, and interpretability of complex analyses. Consequently, human creativity is amplified with trustworthy machine intelligence, resulting in breakthroughs that neither could achieve independently.

It is widely recognized in the literature that traditional AI systems sustain innovation (Truong and Papagiannidis, 2022), especially by enhancing knowledge management through automatic retrieval and processing of vast amounts of unstructured data (Botega et al., 2020; Jarrahi et al., 2023; Mancuso, Petruzzelli, & Panniello, 2024). However, there is a significant gap in understanding how knowledge management is uniquely shaped by XAI within the innovation process. Indeed, we know that XAI provides insights into how and why decisions or predictions are made (Sadeghi et al., 2024), making it potentially suited to support human decision-makers in synthesizing, validating, and applying knowledge (Leoni et al., 2024). However, the current literature on XAI mostly focuses on the impact of XAI on user experience (Kim et al., 2023), information systems management (Meske et al., 2022; Minh et al., 2022), operational decision making, and business applications (Tchuente et al., 2024). While these studies highlight the benefits of XAI, they do not specifically address its role in transforming the way knowledge is managed to drive innovation processes. Therefore, in this paper we study the impact of XAI on innovation by investigating the specific way through which XAI enhances knowledge management during the innovation process. Specifically, the research question addressed in this paper is the following: *"How does XAI influence knowledge management along the four creative phases of the innovation process?"*.

To answer our research question, we conducted a single case study with an embedded logic, following Yin (2014) and Scholz and Tietje (2002). This approach is useful when multiple units of analysis are involved within a single case (Yin, 2014). Indeed, this methodology allows for a more detailed examination of complex phenomena by exploring different subunits within the case (Scholz and Tietje, 2002). Accordingly, we analyzed the case of Intellico, a company specializing in XAI solutions for knowledge management and innovation processes. Intellico is a leading company specializing in the development of XAI solutions tailored for knowledge-intensive industries, where innovation relies heavily on the effective management of complex knowledge. Its core mission is to bridge the gap between advanced AI technologies and human decision-making by providing transparent, interpretable models that empower users to make informed, confident choices. Using an embedded case study approach, we analyzed multiple projects within Intellico to gain a comprehensive understanding of how XAI can transform knowledge management into a driver of innovation.

The primary output of our research is a model that describes how XAI influences knowledge management within the innovation process. Specifically, this model illustrates how XAI sustains manage knowledge along the four creative phases of innovation, namely preparation, incubation, illumination, and verification (Botega et al., 2020).

The study contributes to the literature on AI and innovation (Truong and Papagiannidis, 2022) by providing one of the first empirical investigations on the role of XAI in triggering innovation processes. Specifically, this manuscript reveals that XAI disrupts the traditional way knowledge is managed along the four creative phases of innovation (Alavi and Leidner, 2001) by means of new and integrated knowledge processes. Additionally, the findings highlight the complementary roles of XAI and humans during innovation, thus extending the current understanding of how human actors and XAI systems collaboratively manage knowledge throughout the innovation process (Leoni et al., 2024).

From a practical perspective, the findings offer actionable insights for organizations seeking to leverage XAI in innovation processes. In this way, the study provides a guide for practitioners in fostering environments where human expertise and XAI outputs can complement each other effectively, ultimately enhancing innovation capabilities.

The article is organized as follows. First, we explore the phenomenon of XAI, its relevance for innovation, and its potential role in knowledge management (Section 2). Next, we outline the methodology employed for the single case study of Intellico, detailing sample, data collection, and data analysis strategies (Section 3). The results and discussion (Section 4) present the findings through a process model that highlights how XAI and humans manage and exchange knowledge along the innovation process. Finally, we conclude by discussing the theoretical and practical implications of the study, along with its limitations and suggestions for future research (Section 5).

2. Theoretical background: innovation processes, knowledge management, and explainable artificial intelligence

This section aims to provide an understanding of the relevance of XAI for innovation by focusing on how it optimizes knowledge management processes. The first paragraph focuses on innovation, showing how knowledge is managed throughout the four creative phases of the innovation process. The second paragraph unearths the role that traditional AI systems have in innovation and illustrates the specific characteristics of XAI that can improve knowledge management during the innovation process. This allows us to explain the rationale behind our research objective.

2.1. Knowledge management in the innovation process

Innovation can be defined as *"the successful implementation of creative ideas within an organization"* (Amabile et al., 1996, p. 1154). Specifically, literature identifies four creative phases in which the innovation process can be structured, namely *preparation*, *incubation*, *illumination*, and *verification* (Botega et al., 2020) . Each phase builds upon a specific knowledge process, namely respectively, *knowledge storage/retrieval*, *knowledge creation*, *knowledge transfer*, and *knowledge application* (Alavi and Leidner, 2001).

The *preparation phase* is the initial stage of innovation that focuses on gathering information on the problem or opportunity which gives birth to innovation (Botega et al., 2020). The goal of this phase is to build a foundation of knowledge by identifying gaps, exploring existing solutions, and examining trends, technologies, or patterns to define the scope of the innovation process (Raghu and Vinze, 2007). To this aim, *knowledge storage and retrieval* are crucial (Alavi and Leidner, 2001). Knowledge is usually stored through systematic research, documentation, trend analysis, and by identifying gaps in current understanding of a phenomenon (Al-Emran et al., 2018). Knowledge retrieval involves accessing the stored information to provide context, background, and insights that shape how the problem or opportunity is framed (Al-Emran et al., 2018).

The *incubation phase* follows the preparation stage and allows ideas to emerge, often in an unconscious or indirect manner (Botega et al., 2020). Indeed, in this phase humans involved in innovation give their minds the space to connect disparate insights and expose themselves to diverse stimuli or environments. The purpose is to foster creative connections via *knowledge creation* (Alavi and Leidner, 2001), which refers to the process by which individuals or organizations generate new knowledge through the combination of existing information, experiences, and cognitive processes (Pinho et al., 2012; Savino et al., 2017). Knowledge creation during incubation involves synthesizing ideas, making novel connections, and fostering creative thinking to address

complex problems or opportunities (Wee and Chua, 2013).

The *illumination phase* represents instead the phase when a new idea, solution, or perspective becomes clear (Botega et al., 2020). This results from the synthesis of information processed during the preparation and incubation phases, leading to a creative and original idea. During this phase, the innovative idea that has emerged is then identified, clarified, and made explicit. To this aim, humans leverage *knowledge transfer* (Alavi and Leidner, 2001), that is the sharing or communication of new insights and creative ideas from one individual or team to others or across organizations (Pinho et al., 2012). This involves translating individual insights into shared understanding to foster collaboration (Raghu and Vinze, 2007).

The *verification phase* is the final stage of the innovation process, focused on testing and refining the idea or solution generated during the illumination phase (Botega et al., 2020). Verification ensures that the idea can address the innovation problem or seize the innovation opportunity in a realistic and sustainable manner. To this aim, the innovative solution is empirically tested, and feedback is collected to refine the solution based on practical evidence. In this phase, the newly created and transferred knowledge is put to the test in practical settings to assess its viability and usefulness (Al-Emran et al., 2018). This process is known as *knowledge application* (Alavi and Leidner, 2001).

The knowledge-related processes that must be managed during the four phases of innovation often require the coordinated involvement of multiple teams to ensure that knowledge is effectively stored/retrieved, created, transferred, and applied. This collaborative effort is essential to avoid the formation of informational silos, where critical knowledge becomes isolated within specific departments or groups, hence hindering its accessibility and utility across the organization for innovation purposes (Pinho et al., 2012). Furthermore, knowledge management processes must contend with the challenge of processing vast amounts of data originating from diverse sources, including internal repositories, external databases, and real-time data streams (Al-Emran et al., 2018). Historically, this has been achieved through manual efforts, supported by different tools, such as document management systems, intranets, and structured workflows (Alavi and Leidner, 2001). However, these approaches are often time-consuming, prone to inefficiencies, and limited in their ability to scale with the increasing complexity of modern knowledge ecosystems (Raghu and Vinze, 2007). Recently, AI has emerged as a tool to address these challenges, offering advanced capabilities for automated information processing, allowing human effort to focus on higher-order cognitive processes, and fostering seamless collaborations across organizational boundaries (Brem et al., 2021; Del Giudice et al., 2023). The ensuing paragraph examines this trend in more detail, illustrating how knowledge can be managed via AI during the four creative phases of the innovation process.

2.2. The role of AI on knowledge management in the innovation process

AI is the discipline of building machines that can reason, learn, and act similarly to humans (Kolbjørnsrud, 2024). AI supports innovation by addressing the limits of human information processing (Truong and Papagiannidis, 2022) and allowing for the systematic exploration of new and more exploratory knowledge domains beyond existing ones (Haefner et al., 2021). Accordingly, *"with the assistance of AI, managers can expand beyond their current knowledge base, enabling them to develop more innovative solutions and identify creative opportunities"* Haefner et al. (2021, p. 3). In particular, AI can sustain the *preparation* phase of innovation by identifying patterns and insights from large datasets, which in turn supports creativity by integrating diverse knowledge domains (Haefner et al., 2021). Specifically, AI can enhance knowledge storage/retrieval through automated classification and organization of diverse types of explicit knowledge that can sustain the discovery and generation of new ideas (Jarrahi et al., 2023). Furthermore, AI can influence the *incubation* phase of innovation by analyzing data to assess feasibility, potential impact, and alignment with strategic goals (Truong and Papagiannidis, 2022). In this way, ideas' evaluation is streamlined, as knowledge is created via AI's self-learning and predictive capabilities that recognize previously unknown patterns (Jarrahi et al., 2023). Then, AI can optimize the *illumination* phase of innovation by improving idea's design and development (Truong and Papagiannidis, 2022). Specifically, AI can assist knowledge transfer by connecting people working on the same issue and facilitating collaborative idea development based on feedback and peer review (Jarrahi et al., 2023). Finally, AI can optimize the *verification* phase of innovation by facilitating rapid prototyping through simulation, thus reducing the time and cost of testing ideas (Haefner et al., 2021). Indeed, AI can enhance situated knowledge application by offering more natural and intuitive system interfaces, such as voice-based assistants and chatbots (Jarrahi et al., 2023).

While AI has significantly enhanced knowledge management activities along the innovation process, it is not immune to challenges. A key limitation lies in the opacity of traditional AI systems, which can hinder trust, interpretability, and accountability in decision-making (Meske et al., 2022), that in turn limits a real collaboration between humans and AI (Kolbjørnsrud, 2024). To cope with this, XAI emerges as a vital advancement, offering methods to make AI's operations and outputs more transparent, interpretable, and trustworthy (Wang and Ding, 2024). The following section explores how XAI can further advance knowledge management during innovation, hence ensuring that AI-driven insights are not only accurate but also comprehensible to human stakeholders.

2.3. From artificial intelligence to explainable artificial intelligence (XAI)

Despite the positive impact that AI can have on innovation (Truong and Papagiannidis, 2022), especially in terms of optimized knowledge management (Botega et al., 2020), two main problems affect the application of traditional AI for knowledge management during innovation. First, AI often obscures its underlying mechanisms (Hamm et al., 2023). Second, AI traditionally presents predictive results that are opaque and difficult to understand for non-technical experts (Haque et al., 2023). This lack of transparency becomes particularly critical in knowledge-intensive processes where it is essential to understand and verify the reasoning behind AI-driven insights to ensure their accuracy, reliability, and applicability to the specific context (Minh et al., 2022). These two problems are idiosyncratic in the so-called *"black-box"* AI models (Tchuente et al., 2024). These AI models leverage highly complex systems, such as deep neural networks or ensemble methods, to determine outputs with excellent predictive performance. However, black-box models lack transparency, as the internal structure and the transformation path of inputs into outputs is opaque to both end-users and technical developers of the algorithms (Hamm et al., 2023). The lack in transparency typical of traditional AI systems can result in two negative and opposing human behaviors towards AI. These are the *"automation bias"* and the *"algorithm aversion"*. Automation bias describes a human tendency to over-rely on automated processes, even if they are obscure or potentially contradictory (Elsbach and Stigliani, 2019). Conversely, algorithm aversion refers to a cognitive bias where individuals distrust or are hesitant to rely on automated systems or algorithms, even when these tools may offer superior performance (Allen

and Choudhury, 2022).

To address the problems of automation bias and algorithm aversion that characterize traditional black-box AI and that can limit its applicability for innovation, researchers, practitioners, and policymakers have unitedly called for a larger and more widespread use of XAI. XAI is a type of AI that provides insights into how a decision or prediction is made and how the resulting actions are executed (Hamm et al., 2023).[1] The interest towards XAI has raised after the EU general data protection regulation (GDPR) stating that humans whose data are collected, processed or stored by automated decision-making systems should have the right to "*obtain an explanation of the decision reached*" (European Commission, 2017). In the practitioner's domain, XAI has been recognized as particularly critical in knowledge-intensive use cases, such as healthcare, finance, and innovation, where decision-making processes must be understandable to non-technical users (e.g., doctors, financial analysts, innovation managers) to ensure their trust and reliance on the system (Haque et al., 2023). The research community is also rapidly embracing the study of XAI, with the number of publications on the topic more than doubling between 2022 and 2023 (Mersha et al., 2024).

Specifically, some studies on XAI focus on understanding how this particular type of AI can address the limitations of traditional black-box AI models for innovation. This is possible via four key dimensions, namely *data explainability*, *model explainability*, *post-hoc explainability*, and *assessment of explanations* (Ali et al., 2023). *Data explainability* focuses on providing insights into the data used to train XAI models, ensuring the data are comprehensible and interpretable for both developers and users through techniques such as feature engineering and standardization (Ali et al., 2023). *Model explainability* pertains to the clarity and interpretability of the XAI model's structure, mechanisms, and decision pathways (Ali et al., 2023). This is possible through visual and textual representations of the decision-making procedure and the algorithm's working principles to process inputs into outputs (Haque et al., 2023). *Post-hoc explainability* is applied after a model has been trained to explain its decisions through visualizations, counterfactual explanations, feature relevance scores, and textual summaries that elucidate the reasoning behind predictions (Ali et al., 2023). To ensure comprehensive and effective explainability, XAI systems not only generate explanations automatically but also allow users to request additional, supplementary information as needed (Haque et al., 2023). This improves humans' control on the AI system and avoids cognitive overload (Allen and Choudhury, 2022; Mersha et al., 2024). Finally, the *assessment of explanations* evaluates the quality and effectiveness of the explanations provided by the XAI system (Ali et al., 2023). This includes measures of clarity, relevance, and completeness, as well as how well the explanations align with users' needs (Haque et al., 2023).

By leveraging these four dimensions, XAI amplifies the opportunities to use traditional black-box models by helping human verify AI's decision-making, improving trust towards AI systems, and motivating managers' adoption of AI for innovation purposes (Hamm et al., 2023; Wang and Ding, 2024). This is possible because the higher level of transparency achieved through XAI encourages humans to consider XAI's output more favorably and incorporate it into their innovation activities (Hamm et al., 2023). Indeed, by enhancing the interpretability of AI-generated insights, XAI enables domain experts to validate findings, share knowledge across teams, and improve decision-making throughout the innovation process. Consequently, recent systematic literature reviews on the interplay of AI and innovation are wondering about the potentiality of explainability of XAI for innovation processes (Gama and Magistretti, 2025; Pietronudo et al., 2022), especially in terms of better knowledge management mechanisms that it has the potential to enable. However, while it is recognized that the role of XAI in innovation processes warrants closer attention because explainability enables better alignment of AI-driven insights with human understanding, making it easier to integrate knowledge into the innovation process in meaningful and actionable ways, *"a common understanding of the underlying theoretical capabilities has become increasingly vague and fraught with ambiguity"* (Gama and Magistretti, 2025, p. 1). In fact, most studies on XAI are conducted from a technical perspective, examining users' preferences for XAI interfaces (Kim et al., 2023), quality criteria of personalized explanations (Meske et al., 2022), the effect of XAI on supply chain decision-making processes (Sadeghi et al., 2023), and the best practices in adoption of XAI in industrial applications (Tchuente et al., 2024). Conversely, *"very few studies measure XAI's impact on domain experts"* such as innovation managers who activate and manage knowledge for innovation purposes (Haque et al., 2023, p. 10). Specifically, studies that examine how XAI influence knowledge management along the innovation process are missing, to the best of our understanding. To bridge this gap, this study aims at unveiling the specific ways through which XAI can influence knowledge management along the four creative phases of the innovation process.

Fig. 1 synthesizes the main topics covered in this section and provides the conceptual framework. Specifically, for each creative phase of innovation, a brief description of the importance of the phase is presented, an illustration of the main knowledge management processes traditionally involved in the phase is provided, and the role that AI plays in supporting the specific phase is specified.

3. Methodology

To understand how XAI influences knowledge management mechanisms along the four creative phases of innovation processes, a qualitative methodology was adopted. This approach is particularly effective for investigating complex and contemporary phenomena, allowing for in-depth exploration through the use of multiple sources of evidence (Yin, 2014). Moreover, it has proven especially useful for identifying the role of AI in organizational transformations (Kannelønning, 2024; Kulkov, 2021).

The analysis was designed and conducted adopting an embedded, single case study approach (Scholz and Tietje, 2002; Yin, 2014).

Traditionally, a single case study involves in-depth exploration of a single unit, treating the case as a whole, without further segmentation. This approach allows researchers to gain a deep understanding of a phenomenon within its real-life context, focusing on the case as a singular entity (Yin, 2014).

In contrast, we adopted a single case study with an embedded logic (Scholz and Tietje, 2002). An embedded case study involves not only the examination of the primary case, but also the analysis of multiple sub-units or components within the case. The focus on these sub-units, which may be specific projects, departments, or aspects of the case, allows for a more detailed and multifaceted exploration of the phenomenon under study. In our case, we selected Intellico, a consultancy firm specialized in designing and delivering XAI solutions, as the primary case. Within this case, we analyzed key projects conducted by Intellico as the subunits of analysis, allowing us to explore both the broader organizational context of Intellico and the specific dynamics of individual projects.

The embedded case study approach proves valuable for the purpose of this research. Indeed, by providing a holistic understanding of the Intellico company while simultaneously delving into critical aspects of its individual consulting projects, we were able to address the research

[1] The term XAI was first introduced by Van Lent et al. (2004). The concept was initially used to describe a component in a serious game called "*Full Spectrum Command*", developed for military training by the U.S. Army. The game simulated military operations, where users issued strategic commands. The XAI module allowed the trainee commander, after completing a mission, to select one of their virtual subordinates and inquire about the reasoning behind their decisions and the actions of the troops. This early application of XAI focused on providing transparent explanations for machine behavior, which was crucial for enhancing the trust and understanding of AI systems in complex and high-stakes scenarios.

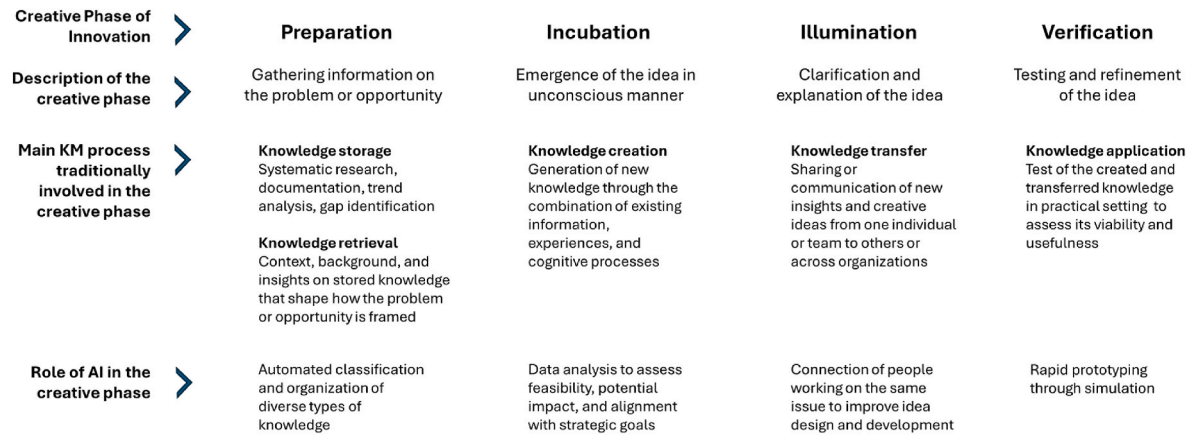


Fig. 1. Phases of innovation processes, knowledge management mechanisms, and the role of AI. Source: Authors' elaboration.

question comprehensively. Additionally, analyzing multiple subunits (i.e., projects) within the same case enhances the reliability of the conclusions through data triangulation from multiple sources, thereby increasing the validity and robustness of the findings (Yin, 2014).

The subsequent sections detail the rationale behind the sample selection, the data collection sources and process, and the method used for data analysis.

3.1. Sample

Intellico (Table 1) is an Italian company specialized in the design and provision of XAI tools and platforms designed to optimize corporate decision-making processes, enhance operational efficiency, and promote transparency in AI systems. Intellico's primary mission is to revolutionize decision-making through XAI, by designing, implementing, and managing XAI platforms tailored for enterprise-level adoption. Specifically, Intellico develops customized applications for smart factory, smart retail, and smart energy, leveraging advanced XAI models for data analytics, demand forecasting, asset management, and process automation.

Intellico represents an ideal case study for this research for several key reasons.

First, its specialization in XAI solutions makes it highly relevant. As a consultancy firm focused on designing and delivering XAI technologies, Intellico's expertise offers a rich context for exploring how XAI influences knowledge management throughout the creative phases of the innovation process.

Second, Intellico's diverse projects provide an opportunity for an embedded case study approach. The firm works on a range of different projects, each applying XAI in different ways. This diversity allows for a deeper exploration of the various sub-units (i.e., individual projects), hence facilitating a more nuanced understanding of how XAI operates across distinct contexts, industries, and challenges.

Third, Intellico's emphasis on innovation and knowledge management is central to its business. Innovation, product design, and technological advancements in XAI are at the core of the firm's operations. Studying Intellico allows for an in-depth investigation of how XAI impacts key knowledge management processes, enhancing the understanding of its role within innovation.

Finally, Intellico offers access to in-depth data, representing a case of convenience sampling. Indeed, Intellico offers privileged access to key stakeholders and project-specific data due to its ongoing collaboration with one of the authors on research initiatives related to other topics. This existing partnership enables a more in-depth exploration through detailed interviews and comprehensive data collection, ensuring the richness and reliability of the insights necessary to address the research question effectively.

Founded in 2020, Intellico combines over 50 years of global expertise in AI and B2B software, thanks to an international team that brings together a unique blend of skills in data science, strategy, finance, and management. The company also benefits from an advisory board comprising leading figures in AI and B2B business development.[2] Since its establishment, Intellico has completed over 25 projects, industrialized four solutions, and developed an XAI platform, i.e., Matilde, that is currently available on the market. Moreover, the company has experienced a steady growth of its economic and financial performance and garnered international recognition. Intellico's 2023 revenue showed a 17 % year-over-year increase, amounting to €427.5K.[3] In terms of recognition, the company has received multiple awards underscoring its significance in the XAI market. In Italy, Intellico won the 2024 Assolombarda Digitalization Award for its advancements in digital innovation and contributions to improve corporate efficiency.[4] Moreover, its XAI platform, named Matilde, was recognized as the "Best App or Digital Platform" at the 2024 Digital Beauty Awards, a Spanish competition focused on digitalization and e-commerce in cosmetics, perfumery, and personal care.[5] Finally, Intellico is among the companies included in ELITE, the pan-European ecosystem of the Euronext Group, which supports small and medium-sized enterprises in scaling and accessing private and public capital markets.[6]

Table 1
Intellico's overview. Source: Authors' elaboration.

Category	Details
Year of Establishment	2020
Size	•20+ employees •50+ clients
Headquarters	Milan, Italy
Business Achievements	•25 completed projects, 4 industrialized solutions, and 1 XAI platform available on the market •Revenue of €427.5K in 2023, marking a 17 % increase compared to 2022
Awards and Recognition	•Winner of the "Digitalization for SMEs" category at the Assolombarda Awards 2024 •Honored at the Digital Beauty Awards 2024 in the "Best App or Digital Platform" category • Member of the ELITE program

[2] https://intellico.ai/who-we-are/.
[3] https://registroaziende.it/azienda/intellico-srl-milano.
[4] https://esgnews.it/social/assolombarda-awards-2024-chi-sono-i-vincitori-e-perche/.
[5] https://digitalbeautyawards.com/desvelados-los-ganadores-de-los-digital-beauty-awards-2024/.
[6] https://www.elite-network.com/it/private-companies/our-companies/intellico-s-r-l.

Intellico develops advanced and customized tools based on XAI that revolve around three key technologies. The first key technology of Intellico is Matilde, an XAI platform allowing companies to systematically analyze, visualize, and interpret model predictions. Knowledge Graphs (KGs) are the second key technology of Intellico. KGs are frameworks that enable the graphical representation of complex relationships and patterns within data, providing an interpretable foundation for informed and strategic decision-making. Finally, the third key technology used by Intellico is Graph Neural Networks (GNNs), an emerging class of machine learning models that support the intuitive visualization and interpretation of AI predictions. By elucidating the logical processes underlying model outcomes, GNNs bridge the gap between machine-generated insights and domain expertise, empowering users to align AI recommendations with their own knowledge base. Matilde, KGs, and GNNs constitute the core of Intellico's technological portfolio and are utilized to design and implement bespoke XAI solutions tailored to the specific needs of diverse industries, including confectionery, manufacturing, and retail. Each deployment involves a collaborative process, starting with an in-depth needs analysis and culminating in the operationalization of tailored platforms. Table 2 provides an overview of Intellico's most significant projects, serving as the basis for the empirical analysis presented in this study.

3.2. Data source

The sources of data used for the study were both primary and secondary sources (Yin, 2014).

Primary data were gathered in the form of semi-structured interviews (Table 3). The interviews, lasting between 40 and 50 min, were recorded and then transcribed to ensure data fidelity for rigorous analysis. Each transcript was emailed to the interviewee for content validation, clarification, and the addition of supplementary information. A total of seven interviews were conducted, involving Intellico's chairman of the board and Intellico's CEO, as well as the main managers responsible for the design and deployment of XAI solutions and a client who benefited from the company's solutions. This approach enabled a more holistic evaluation of Intellico, providing insights into the implemented solutions, the critical activities related to knowledge management, and the impact of XAI services on innovation processes.

The employed interview protocol was structured into five main sections. First, it explored Intellico's resources, capabilities, clients, and offerings to understand the company's business model and operating processed. Second, it delved into the development of XAI applications for the client companies, focusing on each innovation project, the client's involvement in the development and testing of the application, and how client knowledge, as well as data and inputs required for the project, were gathered and managed. Third, our attention was directed to the application of XAI solutions, examining how these solutions facilitated and influenced knowledge management within the client company. Fourth, the questions focused on Intellico's support to its clients in integrating XAI with human efforts, emphasizing techniques to make results comprehensible and useable by end users. Fifth, the protocol asked for challenges and opportunities in the single projects, as well as for themes deemed relevant for the interviewer that were not sufficiently deepened. The full interview protocol is described in Appendix A.

Alongside direct interviews, the research was enriched with additional primary and secondary data (Table 4). Five presentations on Intellico and its offerings were retrieved by the company, while material from the official corporate website served as a primary narrative source,

Table 2
Intellico's projects overview. Source: Authors' elaboration.

Project Name	Sector	Project Scope	XAI technology adopted
Alfa	Food	Optimization of R&D process to support formulation activities	Matilde platform
Beta	Manufacturing and Chemical	Optimization of component reuse in design processes through geometric identification	Knowledge Graph
Gamma	Food	Energy and maintenance optimization	Machine learning platform; Knowledge Graph
Delta	Organic/Km 0 Retail	Optimization of demand forecasting and inventory management	Matilde platform; Graph Neural Networks

Table 3
Interviews details. Source: Authors' elaboration.

ID Interview	Date	Interviewer Role	Main Topic Emerged
ID1	13/05/2024	Chairman of the Board	- Alignment of the study with Intellico's mission and core values
ID2	22/05/2024	Chairman of the Board	- Strategic and technological vision of Intellico - Competitive positioning compared to key market players
ID3	20/06/2024	CEO of Intellico	- Importance of XAI technology within Intellico's overall strategic framework - Overview of Intellico's offering with reference to XAI technology
ID4	11/07/2024	CEO of Intellico	- Prioritization of Intellico's projects that best align with the scope of the study - Identification internal personnel best suited to support the study's data collection - Suggestion of clients that could provide valuable insights for the study
ID5	01/08/2024	Head of Solutions in Intellico	- Intellico's technological portfolio - Overview of Intellico's projects
ID6	24/09/2024	Product Owner & Business Analyst in Intellico	- Requirements gathering and design process - Intellico's product development lifecycle - User-centered design and usability of Intellico's solutions - Metrics and performance of Intellico's solutions
ID7	06/11/2024	R&D Manager of Intellico's client for Alfa project	- Experience on Intellico's solution implementation - Perceived value and impact of the project - Challenges in implementation

Table 4
Primary and secondary data details. Source: Authors' elaboration.

Data Type	Description
Official material on the company	5 presentations on Intellico and its offering
Official website	19 webpages linked to the official website (https://intellico.ai/)
Official blog	28 post on the official blog section (https://intellico.ai/business-cases-and-research-insights/)
Official social media	116 post and 4 videos on the official LinkedIn page (https://www.linkedin.com/company/intellico-ai/)
Articles on magazines	26 articles on business and economics magazines (e.g., Il Sole 24 Ore, Affari Italiani, Industria Italiana, Innovation Post, Assolombarda, Smau, Startup Magazine, Industry4Business, Fabbrica Intelligente)
Video material	118 min (live events, podcast, webinars on YouTube)
Scientific article	1 conference paper on Intellico's algorithms (Dileo et al., 2024)

hence supporting all other data. Notably, the presence of a blog with specific articles allowed for a deeper exploration of Intellico's projects leveraging XAI. The blog discusses the advanced technologies used in individual projects, such as Matilde, KGs, and GNNs, and provides news and insights on XAI tools for sectors like energy, manufacturing, and retail. A total of 28 articles categorized thematically were analyzed as of January 2025.

In parallel, Intellico promotes its solutions and shares content related to its offerings via its official LinkedIn page, which served as an additional data source. Specifically, as of January 2025 the LinkedIn page included four videos explaining the functionalities of Intellico's core technologies and 116 posts featuring job offerings, articles on sponsorships for events and conferences, images of news articles showcasing Intellico's achievements, and explanations of the technological and human pillars underpinning the company.

Alongside material directly provided by the company, third-party sources were also collected. A total of 26 articles from business and economics magazines and 118 min of live events, podcasts, webinars available on YouTube were reviewed. These sources discuss Intellico's activities, solutions, and projects. Complementing these resources, a scientific article was analyzed, which provided a technical perspective on the application of Intellico's advanced technologies for optimizing rapid industrial product development.

3.3. Data collection

Data collection was structured into three steps and designed to explore two central aspects of the research question, namely the knowledge management and the innovation process (Table 5).

The first phase of data collection focused on gathering information related to knowledge management within Intellico's projects. This included examining the processes and tools used to acquire, organize, and share relevant knowledge. The sources used in this phase were primarily internal. Data shared by Intellico and available on the corporate website provided a direct view of its knowledge management processes and practices. Additionally, an interview with Intellico's Head of Solutions and the scientific article on Intellico offered valuable insights into how knowledge is managed through the XAI solutions.

Following the analysis on knowledge management processes, the focus shifted to how this knowledge is used to foster innovation. In this phase, the interview with Intellico's Product Owner & Business Analyst as well as the interview with a client proved particularly valuable, providing detailed information on the development of XAI applications, from initial client engagement to development and implementation. These insights were crucial in understanding how the four creative phases of the innovation process were managed. The company blog was also a key resource for understanding how Intellico has driven innovation through XAI, offering concrete examples of how this technology is applied in projects.

In the third step, the amount of data collected for each knowledge management process and innovation phase was enriched to enable triangulation from multiple secondary sources and refine the constructs of the study. This approach enhanced robustness, increased accuracy, and reduced the risk of bias (Yin, 2014) in the analysis of each project. The additional data for this phase consisted of third-party articles from business and industry magazines, videos and webinars, and LinkedIn posts.

Table 5
Data collection phases. Source: Authors' elaboration.

Data Collection Phase	Data Source	Result
1. Data collection on knowledge management	Presentation shared by Intellico Corporate website Interview ID5 Scientific article	Knowledge management processes
2. Data collection on innovation process	Interview ID2 Interview ID3 Interview ID4 Interview ID6 Interview ID7 Corporate blog	Four creative phases of innovation
3. Enrichment in data collection	Third-party articles Videos and webinars LinkedIn posts	Refinement of knowledge management processes alongside the four creative phases of innovation

3.4. Data analysis

Data analysis was conducted following the Gioia methodology (Gioia et al., 2013).

The Gioia methodology, widely recognized for its rigorous and systematic approach to qualitative research, emphasizes the importance of grounding theoretical insights in empirical data by systematically organizing raw data into three analytical layers, namely first-order categories, second-order themes, and aggregate dimensions. First-order categories emerge directly from companies' data and participants' language, capturing the practical experiences and perceptions of a phenomenon without immediate interpretation. These first-order categories are then abstracted into second-order themes that reflect theoretical constructs or patterns. To this aim, raw data are interpreted in the light of the most appropriate theoretical lens at the base of the study. Finally, the second-order themes are synthesized into aggregate dimensions, representing the overarching conceptual framework. This step-by-step process ensures a robust connection between the data and the emerging theoretical insights.

In this study, the Gioia methodology was employed to analyze how XAI influences knowledge management along the four creative phases of innovation (Fig. 2). Specifically, the first-order categories for this study represent specific actions, tools, and experiences reported by Intellico's team, clients, and external stakeholders. The second-order themes were derived through a rigorous process in which the authors, leveraging their extensive experience in knowledge management, systematically reviewed the relevant literature in knowledge management and cross-referenced it with the empirical data. This approach ensured that the second-order categories were theoretically grounded while also reflecting the patterns emerging from the raw data. As a result, 6 s-order themes were identified, each representing a fundamental knowledge management process. First, there is knowledge acquisition, i.e., the process of collecting and sourcing data from various channels to build an initial knowledge base (Pinho et al., 2012). Second, there is knowledge embodiment, i.e., the integration of acquired knowledge into technical tools, systems, and organizational processes (Demarest, 1997). Third, there is knowledge creation, i.e., the generation of novel insights and ideas by synthesizing and recombining existing knowledge (Wee and Chua, 2013). Fourth, there is knowledge application, i.e., the process of translating knowledge into practice by implementing it in real-world settings (Alavi and Leidner, 2001). Fifth, there is knowledge exploitation, i.e., the process of optimizing and leveraging knowledge to maximize its value and strategic impact (Demarest, 1997; Truong and Papagiannidis, 2022). Sixth, there is knowledge storage and protection, i.e., the process of preserving knowledge assets and ensuring their secure and effective use for future decision-making (Raghu and Vinze, 2007).

Finally, the 6 s-order themes were consolidated into a single overarching dimension, following the approach used in previous studies that structured case study analyses around a unified aggregate dimension (e.g., Mancuso et al., 2023; Mancuso, Petruzzelli, Panniello, Frattini, & Del Giudice, 2024). The single aggregate dimension consists in the role of knowledge management along the four creative phases of innovation. This aggregate dimension was derived by carefully integrating the second-order themes (i.e., the knowledge management processes sustained by XAI) into a unified construct that aligns with the four creative

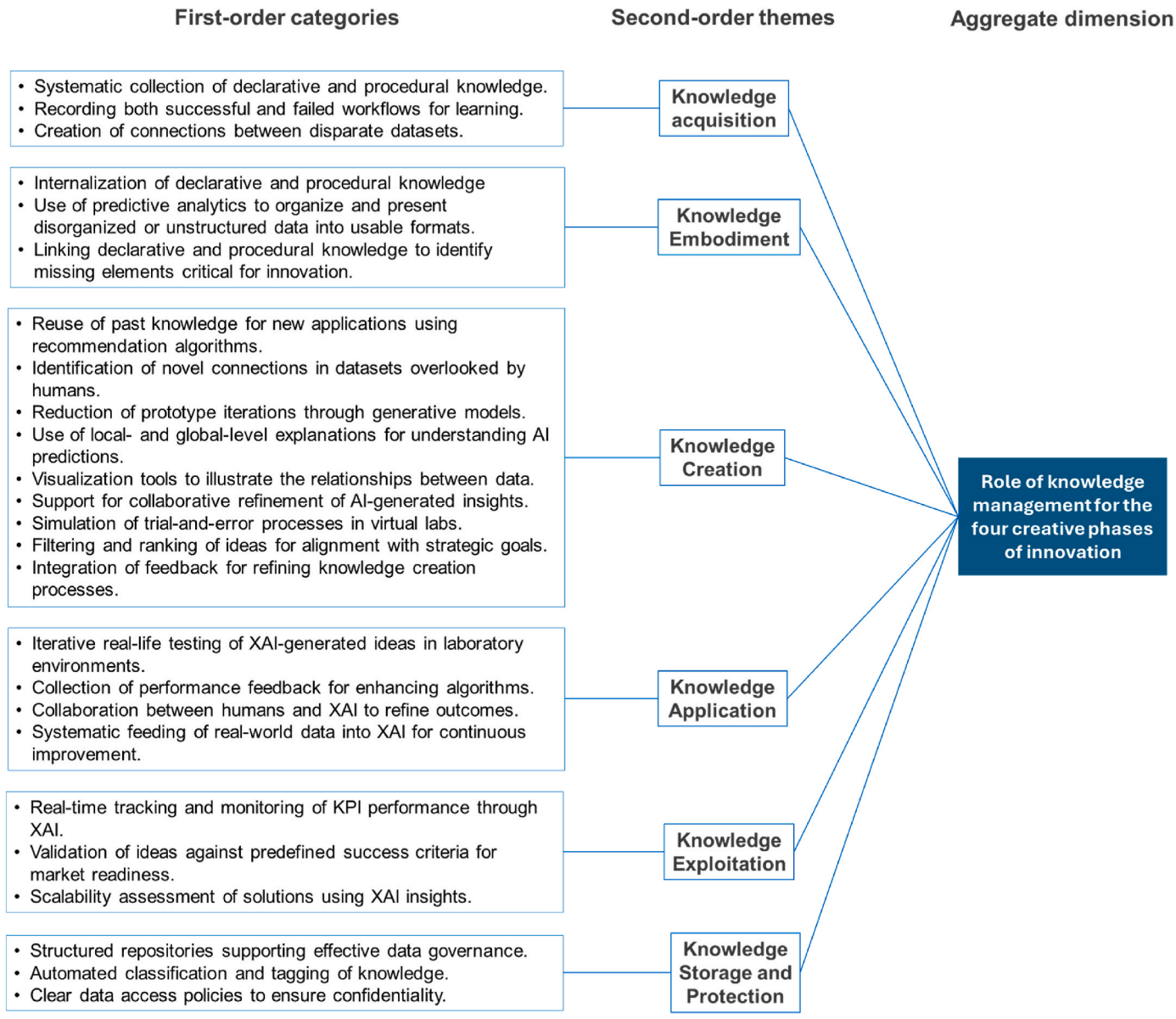


Fig. 2. Gioia methodology applied to the study. Source: Authors' elaboration.

phases of innovation. In particular, each knowledge management process was linked to a specific phase of innovation. Knowledge acquisition and knowledge embodiment were associated with the preparation phase of innovation. During this phase, XAI aids in acquiring and organizing vast amounts of data, ensuring that the innovation process is well-informed and data-driven from the outset. Knowledge creation was connected to the incubation phase of innovation. Indeed, XAI accelerates the generation of new ideas by analyzing existing data, identifying patterns, and producing actionable insights that form the foundation for novel concept development. Knowledge application was linked to the illumination phase of innovation. Specifically, XAI works alongside humans to test and refine ideas in real-world settings, thus supporting users in applying knowledge effectively and adjusting their approaches during the innovation process. Knowledge exploitation was associated with the verification phase of innovation. Indeed, XAI assists in the optimization of knowledge for market readiness by monitoring KPIs and suggesting actionable adjustments. Finally, knowledge storage and protection are associated with all four phases of the innovation process, because safeguarding and preserving knowledge are crucial at every stage. In the preparation phase, it ensures that data and insights are securely stored for later use. During incubation, protecting the intellectual property and refined ideas is vital to prevent premature exposure. In the illumination phase, storing tested knowledge and feedback helps refine ideas further. Finally, in the verification phase, ensuring that the knowledge is safely utilized during implementation is key to maintaining the strategic value of the innovation.

The data analysis was conducted collaboratively by all four authors. Two of the authors conducted the interviews, with one, as an expert in qualitative research methodology, led the first-order coding. The second-order codes and aggregate dimension were developed through a joint effort, with input from all the four authors, who specialize in knowledge management, innovation management and artificial intelligence. This collaborative approach ensured a robust and comprehensive analysis, integrating both methodological rigor and domain expertise.

4. Findings

The main outcome of this study is a model delineating how XAI influences knowledge management along the four creative phases of innovation (see Fig. 3). In the preparation phase, XAI facilitates

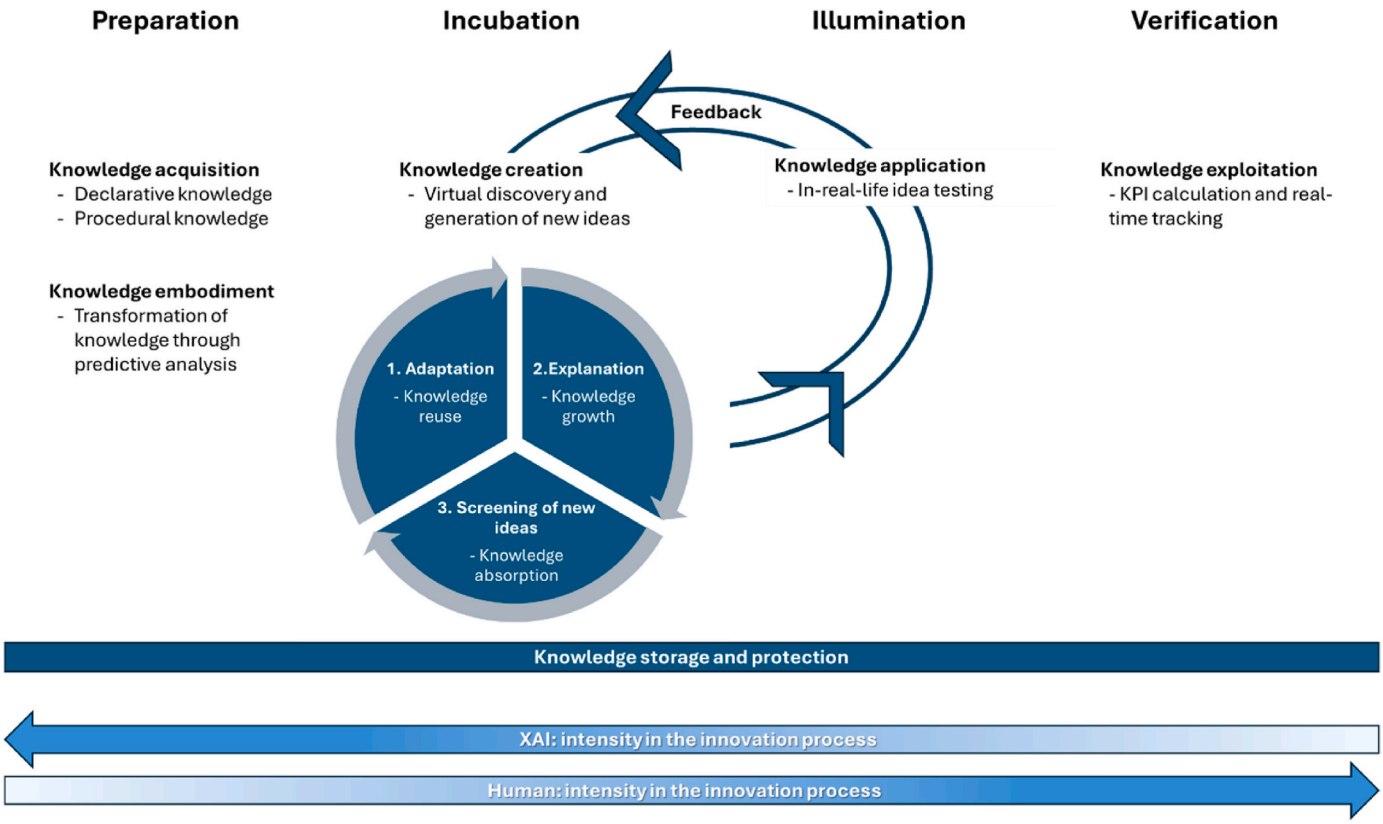


Fig. 3. Model of XAI's influence on knowledge management along the four creative phases of innovation. Source: Authors' elaboration.

knowledge acquisition and knowledge embodiment, enabling teams to effectively gather, organize, and internalize insights relevant for innovation. During the incubation phase, XAI supports knowledge creation through a dynamic cycle of adaptation, explanation, and screening of new ideas. This process involves knowledge reuse, knowledge growth, and knowledge absorption. In the illumination phase, the focus shifts to knowledge application, where humans test XAI-generated ideas in real contexts, providing feedback to improve and extend the system's algorithms. In the verification phase, humans monitor a set of Key Performance Indicators (KPIs) to validate and refine the ideas generated and applied through XAI's support, ensuring knowledge exploitation through the development of solutions for commercialization. Throughout all these phases, there is a continuous emphasis on knowledge storage and protection to ensure that the data managed via XAI are securely preserved and accessible for future use. The following discussion delves deeper into each phase of the innovation process, highlighting the specific mechanisms by which XAI enhances knowledge management.

4.1. Preparation phase: knowledge acquisition and knowledge embodiment

In the preparation phase of the innovation process, knowledge acquisition and knowledge embodiment are the main knowledge processes influenced and triggered by XAI.

Knowledge acquisition refers to the systematic collection and assimilation of information from diverse sources (Pinho et al., 2012). This activity *"necessarily starts with a company's data foundation: while infinite historical data is not always required, a minimum of valid data is indispensable"* [7] (CEO of Intellico). It becomes clear that *"a critical activity is data retrieval, because not all companies can correctly track information, although using an XAI-based approach helps more than a traditional one"* (Head of Solutions at Intellico). In fact, XAI facilitates knowledge acquisition by selecting, collecting, organizing, and mapping both declarative knowledge and procedural knowledge (Fowler, 2000). Declarative knowledge encompasses facts, information, and static content such as research findings, technical specifications, and organizational data. Declarative knowledge is automatically retrieved by XAI via manual registration or by exploiting the XAI platform Matilde for both automatic registration and data exportation from other systems. Procedural knowledge focuses on the methods and processes for performing specific tasks, such as production sequences or steps for troubleshooting technical issues. XAI acquires procedural knowledge through analysis of workflows, task sequences, and historical actions, learning how tasks are typically completed in a given domain. Specifically, regarding the acquisition of procedural knowledge, it is useful to *"record not only successful sequences but also failed experiments, because they help models consider them as negative cases"* (Product Owner & Business Analyst at Intellico). Therefore, the use of XAI offers a significant advantage for knowledge acquisition by enabling the incorporation of knowledge about failures, an aspect often overlooked in traditional (i.e., non XAI-led) knowledge acquisition processes. Indeed, traditional methods for knowledge acquisition often leave procedural failures unaddressed due to the lack of transparency, traceability, and contextual explanations within the system. Without these elements, it becomes challenging to identify the root cause of the problem underlying the failure, as traditional approaches do not provide the insights needed to trace errors back to their procedural origins or understand the interconnected factors contributing to failures. In contrast, XAI offers a structured understanding of why certain procedures led to suboptimal outcomes, offering actionable insights to refine and improve the missteps, inefficiencies, or errors embedded in the processes. As remarked by an interviewed: *"XAI's ability to systematically capture and explain procedural failures*

[7] https://www.youtube.com/watch?v=pFsF7RxvNa8.

empowers domain experts to analyze these issues in real-time and incorporate the lessons learned into their workflows" (Product Owner & Business Analyst at Intellico). By acquiring both declarative and procedural knowledge, XAI broadens the scope and depth of information available for innovation, enabling a more comprehensive preparation phase. However, *"it is not guaranteed that having a lot of data means having the data that is truly useful for the innovation process"* (R&D Manager of Intellico's client for the Alfa project). In this sense, XAI supports the preparation phase of innovation through the process of knowledge embodiment.

Knowledge embodiment is the process by which acquired knowledge is internalized and transformed into actionable insights (Demarest, 1997). XAI plays a pivotal role here through predictive analytics, which process and model the acquired data to forecast future scenarios and identify potential opportunities by providing interpretable explanations of the factors driving these forecasts. Unlike traditional AI, which often operates as a "black box", XAI enhances human understanding by uncovering the "why" and "how" behind its predictions. As highlighted by the CEO of Intellico, XAI *"provides not only predictions but also explanations of the drivers that impacted the prediction […] helping managers better understand the suggestions provided by algorithms while maintaining human control"*.[8] This interpretability ensures that decision-makers are equipped with actionable insights while retaining the ability to critically evaluate and refine outputs. Indeed, *"data can be very disorganized or unstructured, and through predictive analytics XAI helps with data preparation and visualization. For many clients this represents a significant achievement because they have never performed such analyses"* (Product Owner & Business Analyst at Intellico). Through predictive analyses enriched with explanations, abstract or fragmented knowledge can be represented in a usable and structured format, aligning closely with the organization's goals. Specifically, XAI's ability to transparently link declarative and procedural knowledge helps identify gaps in product development while providing a clearer pathway for addressing them. This mechanism is crucial for the subsequent incubation phase.

4.2. Incubation phase: knowledge creation

In the incubation phase of the innovation process, knowledge creation is a critical task. Traditionally, knowledge creation is defined as *"the development of new ideas through human interactions of explicit and tacit knowledge"* (Wee and Chua, 2013, p. 960). We find that XAI plays a pivotal role in knowledge creation through a three-step cyclic model made of adaptation, explanation, and screening of new ideas. This three-step cyclic model decreases reliance on human interactions for knowledge creation, by involving virtual discovery and generation of new ideas.

The first step of knowledge creation, i.e., adaptation, involves adjusting extant knowledge to novel and unexpected contexts (Andriani et al., 2017). XAI systems facilitate adaptation through knowledge reuse, allowing large datasets to be leveraged, uncovering and clearly illustrating connections or patterns that humans might overlook. Knowledge reuse is the process of using knowledge from past experiences or projects for new applications, reducing duplication of efforts and avoiding similar mistakes (Demarest, 1997; Wee and Chua, 2013). XAI promotes knowledge reuse to drive adaptation in specific ways. In particular, by synthesizing and repurposing insights via recommendation algorithms and generative models, XAI *"suggests new experimental approaches to reduce the number of prototypes needed"* (R&D Manager of Intellico's client for the Alfa project). In this way, XAI ensures that declarative and procedural knowledge can be reused innovatively. Moreover, XAI's ability to provide clear explanations of its processes and outputs fosters a higher level of trust in its interactions with human decision-makers. By making its reasoning transparent, XAI not only suggests new applications but also helps domain experts understand why specific recommendations emerge. This transparency reduces uncertainty and enhances the confidence of human collaborators in relying on XAI-driven insights. As summarized in a technical paper on Intellico's solution, XAI *"can suggest a new product design, given the past trials and the desired consumer-appealing characteristics"* (Dileo et al., 2024, p. 5), while ensuring that the rationale behind these suggestions is explicitly communicated. This combination of innovative reuse and enhanced trust underscores XAI's transformative role in knowledge adaptation.

The second step of knowledge creation, i.e., explanation, focuses on a better understanding of ideas through in-depth exploration of insights (Wang and Ding, 2024). The Head of Solutions at Intellico precisely explains XAI's added value: *"XAI allows us to shape reliable AI solutions capable of explaining the predictions they process. We can understand the elaborated path and what the network has learned to arrive at a particular conclusion"*.[9] Specifically, as noted by Dileo et al. (2024), predictions and explanations are provided within the XAI platform Matilde, which includes data exploration, visualization, and editing modules. Within this platform, two types of explanations can be accessed, defined as local-level and global-level explanations. *"Global-level explanations are useful for identifying qualitatively the most important characteristics for obtaining a desired product property"*, while *"local-level explanations are useful for identifying potential starting recipes – nodes – for new formulation trials to test in our platform or real labs, as well as product characteristics – features – that impact the value of the target property"* (Dileo et al., 2024, p. 8). Through the generation of global-level and local-level explanations, XAI supports knowledge growth, which refers to the expansion, enhancement, and refinement of an individual's or group's knowledge base (Ruggles, 1998). XAI supports knowledge growth by generating interpretable information into how conclusions are derived. Specifically, visual or textual explanations illustrate how specific inputs lead to particular predictions or recommendations. As the Head of Solutions at Intellico declared: *"XAI is represented through graphs, where each point represents a recipe, and the closer the points, the more similar the recipes. The Explain function helps us understand how to approach a particular graph/-recipe by selecting the recipe to mimic, and the model will indicate the variables that can bring you closer to it"*. This transparency enables human experts to build upon AI-generated insights, challenge assumptions, and collaboratively refine ideas. Indeed, as *"the Explainer shows the most important product characteristics for a certain expected property and the most important nodes, i.e., previously tested recipes, for reaching the desired characteristic"* it *"allows the R&D department to visualize the obtained graph for investigating new possible trials"* (Dileo et al., 2024, p. 6). *"This is how human-machine collaboration is born, as observing the prediction and the model's key elements enables improvement"*, specifically allowing for *"identifying new recipes for reaching a desired property"* (Dileo et al., 2024, p. 14).

The third step of knowledge creation, i.e., screening of new ideas, involves reviewing and selecting the most innovative ones for further exploration (Truong and Papagiannidis, 2022). XAI supports this process by streamlining knowledge absorption. Knowledge absorption refers to the processes through which individuals or teams internalize and integrate new knowledge to enhance their understanding, skills, and capabilities (Sjödin et al., 2019). XAI streamlines knowledge absorption by filtering relevant knowledge, connecting it with prior understanding, and embedding it into actions and decisions, ensuring that the knowledge can be effectively utilized for practical applications. The Head of Solutions at Intellico clarifies how knowledge absorption, crucial for screening new ideas, occurs through XAI within the Matilde platform:

[8] https://www.youtube.com/watch?v=pFsF7RxvNa8.

[9] https://www.industriaitaliana.it/cluster-fabbrica-intelligente-knobs-sfridoo-intellico-start-up/.

"The screening of new ideas is based on a trial-and-error process, like a recipe, that is modifying an ingredient or element, adding another, and understanding the impact on the final product. With the XAI platform Matilde, the company engages in this core activity with the aim of simulating this trial-and-error operation in a virtual laboratory, consulting the recipe history, modifying ingredients, and observing their effects on the product. Thus, XAI integrates into the core of the innovation process, so that trial and error is no longer conducted physically, consuming time for testing, but is improved by keeping track of all tests and saving them in the system to enhance knowledge absorption of past experiences". Beyond these technical capabilities, XAI fosters trust in the screening process by ensuring that the logic behind decisions is fully traceable and auditable. *"This transparency enables human users to understand why certain ideas are highlighted as viable while others are not"* (Head of Solutions at Intellico). In particular, trust is reinforced by XAI's ability to provide comparative analyses, which demonstrate how different ideas generated within the system are clustered and ranked against predefined criteria, such as feasibility, potential impact, and alignment with strategic goals. By generating detailed explanations alongside its virtually-conducted trial-and-error process, XAI highlights the most viable ideas while providing additional declarative knowledge, such as why specific ideas meet or exceed the set criteria and how they align with broader objectives. The additional declarative knowledge provided by XAI allows domain experts to better understand the reasoning behind the system's recommendations, enhancing confidence in the screening process. By allowing users to see not only which ideas are prioritized but also the specific factors driving their ranking, XAI strengthens the relationship between humans and technology during ideas' screening, ensuring a high level of human trust in interpreting the system's outputs.

Together, the three steps of adaptation, explanation, and screening of new ideas form a continuous loop of knowledge reuse, knowledge growth, and knowledge absorption, empowering organizations in the incubation phase to create the knowledge that will be applied in the illumination phase.

4.3. Illumination phase: knowledge application

In the illumination phase of innovation, the ideas generated and refined by the XAI system undergo real-life testing. This phase marks the critical transition from abstract concepts to tangible applications, where the feasibility, functionality, and value of ideas are assessed in real-world scenarios. As explained by the Head of Solutions at Intellico: *"It is an iterative process where each hypothesis is tested and verified, and additional data is collected in the lab. For example, since in many cases more than one parameter needs to be predicted, a first round of virtual simulation is conducted. The model provides a hypothesis for an improved recipe, the R&D team examines the structure of the ingredients, and tests are performed in the lab to gather the necessary parameters. XAI can be used iteratively, as new information is gathered in the field, parameters are measured, a more precise value is obtained, and the model is updated to recalculate the data"* (Head of Solutions, Intellico).

> Through this process, knowledge application, i.e., the process of using created knowledge to solve problems, make decisions, or perform tasks in practical, real-world scenarios (Alavi and Leidner, 2001), is activated. Specifically, knowledge application is sustained via interaction between humans and XAI systems, as the Head of Solutions elaborates: *"Using an XAI-based approach is more beneficial compared to a traditional one, where information is taken, a result is developed, and if it's wrong, it's discarded. On the other hand, XAI analyzes and develops hypotheses, simulates them in a virtual lab, and provides statistics on both what worked and what didn't, offering recommendations on how humans can improve the real lab test. This fosters human-machine collaboration: by observing the prediction and key elements of the model, it can be improved. [….] XAI provides the R&D team with a direction to move toward; instead of relying solely on experience, as in the traditional process, the XAI approach - through iterative cycles of refinement and in a virtual manner - guides the application of human knowledge and enhances it"* (Head of Solutions, Intellico).

During real-life testing, R&D managers implement XAI-generated ideas in practical settings, generating valuable feedback about their effectiveness and suitability, especially in terms of the *"discrepancy between the actual value in the lab and the value predicted by the XAI model"* (Product Owner & Business Analyst, Intellico). Specifically, success metrics and observed challenges are systematically fed back into the XAI system to improve the immediate solution, enhance the XAI algorithm, and refine the knowledge creation process. *"This feedback loop allows the system to better identify and explain the factors influencing the prediction"* (Product Owner & Business Analyst, Intellico). In this way, the feedback loop ensures the model adapts and evolves to better align with real-world conditions, promoting continuous improvement and making the knowledge creation process more robust over time. As the Head of Solutions at Intellico declared: *"The recipe that, according to the virtual simulations of the XAI model, aligns best with the objective is selected. Then, it is brought to the lab, tested, and the initial test data are used to determine if the results meet expectations. If not, this information is provided to the model as new training instances for updating the trained model and generating a new recipe"* (Head of Solutions, Intellico).

4.4. Verification phase: knowledge exploitation

In the verification phase, successfully tested ideas are validated to ensure they meet the standards required for commercialization. This phase involves assessing the feasibility, performance, and market readiness of the innovation. The central process in this phase is knowledge exploitation, which includes gathering insights on the innovation's effectiveness, identifying potential improvements, and aligning it with market needs (Demarest, 1997; Truong and Papagiannidis, 2022). XAI plays a pivotal role in this phase by facilitating KPI calculation and real-time tracking. This concept is reinforced by the CEO of Intellico, who stated: *"XAI enables real-time KPI monitoring, ensuring the scalability of solutions"*[10] (CEO of Intellico). For example, in the Delta project, KPIs for demand forecasting in the retail sale of organic products included forecast granularity and accuracy, accounting for operational parameters (e.g., stock levels) and the distinct characteristics of different product categories (e.g., fresh vs. long-shelf-life products). It was demonstrated that applying the algorithm improved these KPIs, resulting in a more effective demand forecasting model.[11] Indeed, through XAI's ability to analyze and present data transparently, stakeholders can better understand how well an innovation performs against predefined success criteria. As highlighted by a participant in the Alfa project: *"The main driver of the project was the optimization of time-to-market processes. Therefore, the general idea was to also identify a KPI that could correspond to a potential return on investment, which is the primary driver behind business decisions today. For instance, in the first part of the project with Intellico (proof of concept), the main KPI was to assess the quality of the model, that is, to determine whether this type of approach could indeed provide an advantage"* (R&D Manager of Intellico's client for Alfa project).

4.5. Knowledge storage and protection across all phases of innovation

Throughout each phase of the innovation process, i.e., preparation, incubation, illumination, and verification, knowledge storage and protection play a critical role in preserving and securing valuable insights and information (Al-Emran et al., 2018). As stated by Intellico's CEO: *"The structured collection of data and digital archiving form the foundation*

[10] https://www.youtube.com/watch?v=pFsF7RxvNa8.

[11] https://intellico.ai/it/blog/caso-di-studio-previsione-della-domanda-per-unazienda-del-retail/.

for effective model training and prediction". Moreover, ensuring compliance with emerging regulations is essential, as highlighted by the Product Owner & Business Analyst of Intellico: *"Adhering to emerging regulations ensures transparency and traceability in algorithmic decision-making"*.[12] This step was also crucial in the Gamma project, where Intellico worked closely with the client to digitalize business processes and enable XAI application via application of regulations on data management.[13] These dual processes of knowledge storage and knowledge protection are fundamental for maintaining the continuity and integrity of the knowledge managed via XAI.

Knowledge storage involves systematically capturing and organizing knowledge in repositories for future use (Raghu and Vinze, 2007). Intellico's CEO emphasized that *"it is essential to* support *XAI models with effective data governance"*.[14] On this, Intellico's Head of Solutions elaborated that *"data governance is the set of policies, procedures, and tools ensuring secure, high-quality, and compliant data management within an organization. It aims to maximize the strategic value of data by ensuring its quality, relevance, and regulatory compliance. It includes tools like business glossaries, data catalogs, and data lineage"*.[15] XAI systems enhance knowledge storage with automated classification and tagging, structuring explicit knowledge (e.g., documents, datasets, and models) in intuitive, searchable ways. Furthermore, XAI systems also maintain detailed records of the insights derived throughout innovation, documenting both declarative and procedural knowledge for future reference.

Knowledge protection focuses on safeguarding intellectual property, sensitive data, and proprietary processes from unauthorized access or misuse. In the context of XAI, this includes implementing robust security measures. As emphasized by an Intellico's client: *"One of the most important aspects of the project was managing confidential data, as it required externally sharing recipe-related information, which is the epitome of intellectual property"* (R&D Manager of Intellico's client for Alfa project). To ensure knowledge protection, clear data access policies must be established. Intellico's Head of Solutions clarified: *"when a dedicated Intellico team* supports *the innovation project for company A, it does not have access to the information of the innovation project of company B, however similar it may be, because these are armored systems at the level of management of the intellectual part"* (Head of Solutions of Intellico). In this way, individuals working on the same innovation issue within the organization are safely connected, so that advanced collaboration is promoted. In fact, *"Intellico technology must be specialized on the context of the client company and the trained model is part of the know-how of the client company, it becomes intellectual property of the client company that cannot be shared with others, and this is why, when a client comes to us for an innovation need, they cannot expect to have a model already trained"* (Head of Solutions of Intellico). In addition to data policies, innovation supported by XAI requires *"advanced systems to guarantee the security and protection of company data"*[16] (CEO of Intellico). *"To ensure this, we define clear access controls, use encryption to protect during storage and transfer, and track the flow of data to ensure its integrity. Additionally, we identify and mitigate risks through policies, security procedures, and vulnerability management techniques"*[17] (Head of Solutions at Intellico).

[12] https://www.youtube.com/watch?v=2-dbhr-ojss.

[13] https://intellico.ai/it/blog/ottimizzazione-energetica-e-manutentiva-intelligenza-artificiale-guida-la-trasformazione-nel-settore-alimentare/.

[14] https://www.youtube.com/watch?v=pFsF7RxvNa8.

[15] https://intellico.ai/it/blog/data-governance-per-la-generative-ai-perche-e-giusto-parlarne-e-agire/.

[16] https://www.youtube.com/watch?v=pFsF7RxvNa8.

[17] https://intellico.ai/it/blog/data-governance-per-la-generative-ai-perche-e-giusto-parlarne-e-agire/.

5. Discussions and conclusions

This paper explores the intersection of XAI and knowledge management within the innovation process. Through the single, embedded case study of Intellico, we developed a model explaining how XAI influences knowledge management along the four creative phases of innovation, namely preparation, incubation, illumination, and verification.

Our model emphasizes how the use of XAI inherently changes the way knowledge is managed along the four creative phases of innovation. Indeed, as remarked in the theoretical background section (see Fig. 1), traditionally each phase of the innovation process has been closely associated with a specific knowledge management process. In particular, the preparation phase has typically been linked to knowledge storage and retrieval, the incubation phase usually revolves around knowledge creation, the illumination phase is traditionally associated to knowledge transfer, and the verification phase normally is reserved for knowledge application (Alavi and Leidner, 2001). Our model builds on these assumptions and unveils how they are challenged and expanded when integrating XAI in the innovation process. In particular, our model emphasizes that the use of XAI during innovation reshapes the traditional one-to-one association between each innovation phase and each knowledge management mechanisms, as synthesized in Table 6. Interestingly, this points to a significant change in the nature of the knowledge management processes and mechanisms along the steps of the innovation process, when the latter is heavily supported through XAI.

The study reveals that in the *preparation phase* the dominant knowledge management processes shift from the traditional knowledge storage and retrieval, where individuals access existing knowledge repositories (Alavi and Leidner, 2001), to knowledge acquisition and knowledge embodiment, where the XAI system systematically collects and transforms information from various sources in actionable insights (Konar, 2018). This shift occurs because XAI acquires vast amounts of previously inaccessible data, including structured internal databases, unstructured external datasets, and real-time information streams (Botega et al., 2020). In addition to automatically acquiring knowledge, XAI also transforms raw data into actionable insights through predictive analytics and sophisticated modeling, allowing for knowledge embodiment (Botega et al., 2020). In this way, XAI can uncover patterns, insights, and trends within data for a broader and deeper understanding of the environment in which innovation is occurring. This is essential in fast-paced, dynamic environments where static knowledge retrieval may fall short (Konar, 2018).

We spot that, in the incubation phase, knowledge creation remains the central knowledge management process in both traditional and XAI-led innovation, involving the generation of new ideas and insights (Alavi and Leidner, 2001). However, our study makes it evident that the nature of the knowledge creation process evolves with the use of XAI. Specifically, traditional knowledge creation relies on human-driven processes like brainstorming, intuition, and iterative collaboration, resulting in high dependence on serendipity (Lane et al., 2021). Moreover, the traditional knowledge creation process is based on the "sources-uses-outcomes approach" (Pinho et al., 2012), which is a linear process with few opportunities to integrate new knowledge dynamically. In contrast to this, we found that XAI introduces a cyclical process of knowledge creation involving knowledge reuse, growth, and absorption. Specifically, we found that each cycle builds on previous iterations, leveraging a systematic and scientific adaptation (Andriani et al., 2017), explanation (Wang and Ding, 2024), and screening of new ideas (Truong and Papagiannidis, 2022). In this way, existing knowledge is repurposed for novel applications in a structured and unlimited manner, which overcomes human cognitive limits, while the serendipitous nature of traditional knowledge creation is enhanced via a systematic procedure (Lane et al., 2021). As a consequence, XAI-knowledge creation shifts from intuition-driven methods to data-driven processes. This shift enables organizations to access a broader range of ideas in a faster, more

Table 6
The role of XAI along the four creative phases on innovation: changes in knowledge management processes. Source: Authors' elaboration.

	Key Knowledge Management Mechanism in Preparation	Key Knowledge Management Mechanism in Incubation	Key Knowledge Management Mechanism in Illumination	Key Knowledge Management Mechanism in Verification
Traditional (non XAI-led) innovation process	Knowledge storage and retrieval	Knowledge creation: linear sources-uses-outcomes approach	Knowledge transfer	Knowledge application
XAI-led innovation process	Knowledge acquisition and embodiment	Knowledge creation: cyclical knowledge reuse, growth, and absorption	Knowledge application	Knowledge exploitation

systematic, and more scalable way.

The article also highlights that, in the *illumination phase,* the core knowledge process managed via XAI is knowledge application (Alavi and Leidner, 2001), which centers on the practical implementation and real-world testing of ideas, rather than the traditional knowledge transfer process (Alavi and Leidner, 2001), where insights and ideas are shared and disseminated across teams to foster collaboration and validation. While traditional knowledge transfer often ends with human interpretation, the use of XAI in the illumination phase shifts the emphasis from general dissemination of conceptual frameworks or theoretical ideas to cost-effective and practical experimentation of ideas (Truong and Papagiannidis, 2022). In this way, XAI provides actionable and specific insights tailored to the immediate needs of the innovation process, ensuring that the knowledge generated is directly applicable and effective in real-world settings. This change reflects a move toward action-driven innovation, where the value of knowledge lies in its ability to be applied and tested in real-world scenarios (Misseyanni et al., 2018).

Finally, we find that in the *verification phase* knowledge exploitation achieved via XAI substitutes the traditional knowledge application process. XAI supports this transition via KPIs calculation and real-time tracking that in turn ensures the scalability of the innovative solution. Indeed, assessing how an innovative solution performs across multiple metrics allows identification of strengths and areas for improvement, facilitating the development of a commercially viable product (Ghasemaghaei and Calic, 2020). The transition from merely applying knowledge to actively exploiting it for commercial success reflects a move toward a more market-centric approach in the final stages of innovation. Indeed, thanks to the use of XAI, innovation is more closely aligned with the market rather than stopping at internal validation (Aydin, 2021).

Below, the main implications for theory and practice, as well as the limitations of the study, are provided.

5.1. Theoretical implications

This study contributes to the XAI, knowledge management, and innovation literatures.

As for the contribution to XAI literature, the study expands the scope of XAI research beyond its traditional focus on technical explainability. Indeed, traditionally the research on XAI has focused on its technical aspects, such as improving explainability or designing algorithms for better human-machine collaboration (Kim et al., 2023; Meske et al., 2022). Conversely, this study approaches XAI from a non-technical perspective and unveils the role of XAI as an enabler of knowledge-intensive processes along the four creative phases of innovation.

Regarding the contribution to the knowledge management literature, we found that the use of XAI inherently changes the way knowledge is managed along the four creative phases of innovation. Indeed, traditionally each stage of the innovation process has been strongly associated with a distinct knowledge management process. Specifically, the preparation phase usually involves the storage and retrieval of knowledge, the incubation phase normally focuses on generating new knowledge, the illumination phase is traditionally linked to the sharing and dissemination of knowledge, and the verification phase in general demonstrates the application of knowledge (Alavi and Leidner, 2001). Our study reveals that the integration of XAI significantly alters these traditional connections between each phase of innovation and each knowledge process. Specifically, we found that in the preparation phase XAI shifts the focus from static knowledge retrieval to dynamic acquisition and embodiment of actionable insights (Botega et al., 2020; Konar, 2018). Then, we found that during the incubation phase XAI replaces intuition-driven knowledge creation with systematic, data-driven cycles of knowledge reuse, growth, and absorption (Andriani et al., 2017; Sjödin et al., 2019; Wang and Ding, 2024). Moreover, we found that in the illumination phase XAI emphasizes practical, cost-effective application of knowledge over its broad dissemination (Truong and Papagiannidis, 2022). Finally, we found that in the verification phase the traditional application of knowledge is substituted via a market-aligned exploitation of knowledge that enhances scalability and commercial viability (Aydin, 2021). With these results we contribute to knowledge management literature (Savino et al., 2017; Botega et al., 2020) by means of a model describing how XAI disrupts traditional knowledge management processes along the four creative phases of innovation.

Regarding the contribution to innovation literature, this research bridges two critical gaps on human-machine collaboration during innovation.

First, previous studies have predominantly focused on how AI collaborates with humans for innovation (Gama and Magistretti, 2025; Truong and Papagiannidis, 2022). These studies have overlooked the unique potential of XAI, which not only performs innovation tasks but also explains the rationale behind its predictions, thus sustaining higher levels of human-machine collaboration. Therefore, we contribute to the literature at the cross-road of innovation and AI by emphasizing how XAI fosters higher trust in human-machine interaction, specifically in creating knowledge. Indeed, in the adaptation phase of knowledge creation, XAI's transparency in explaining how knowledge is repurposed from past experiences enables human users to confidently trust its recommendations, knowing the reasoning behind them. In the explanation phase of knowledge creation, XAI enhances trust by providing interpretable, actionable insights through local and global explanations, allowing human experts to understand the system's decision-making process and refine their strategies accordingly. Finally, during the phase of knowledge creation revolving on the screening of new ideas, XAI's ability to justify why certain ideas are prioritized over others strengthens trust by ensuring the decision-making process is fully auditable and transparent. With these outcomes, we defined three specific ways through which XAI can be considered a more reliable partner in decision-making as compared to traditional AI, improving trust and collaboration between humans and machines.

Second, our study fills a critical gap in understanding the roles of XAI and humans in fostering innovation (Gama and Magistretti, 2025; Haque et al., 2023). Specifically, the study suggests that XAI's contribution to innovation is definitely intense during the early stages of innovation (i.e., preparation and incubation). Here, XAI represents the main innovation actor as it allows for gathering of information, adaptation, explanation, and the screening of new ideas. In contrast, the findings indicate that humans' contributions to innovation is higher in the latter stages of

innovation (i.e., illumination and verifications), where humans are the primary innovation actors and are supported by XAI in testing new ideas and monitoring KPIs. In particular, we noted that in the preparation phase XAI replaces the extensive human effort required to identify, extract, and interpret relevant knowledge, thereby reducing cognitive and analytical burdens (Garbuio and Lin, 2021). Moreover, we note that in the incubation phase XAI's automation in idea screening accelerates the discovery of new solutions (Verganti et al., 2020), while its enhanced explanatory capabilities allow humans to better understand, validate, and build upon the system's findings, thus overcoming cognitive limitations (Garbuio and Lin, 2021). Then, we found that while XAI provides insights and suggestions in the illumination phase, humans play a critical role in experimenting with these ideas in real-world contexts. Their observations and refinements not only validate the system's outputs but also contribute to its learning, creating a synergistic relationship between humans and machines (Stiennon et al., 2020). With these results, the manuscript contributes to innovation literature by underscoring the distinct roles of XAI and humans during innovation (Kolbjørnsrud, 2024).

5.2. Managerial implications

This study offers several actionable insights for innovation managers and policymakers seeking to leverage XAI to optimize knowledge management during the innovation process.

First, the paper underscores that managers can utilize XAI to support decision-making at different stages of the innovation process. In the preparation phase, XAI enables informed knowledge acquisition and embodiment by analyzing diverse data sources. During incubation and illumination, XAI facilitates the discovery of innovative ideas and their practical testing. Finally, in the verification phase, XAI sustains calculation and tracking of KPIs to guide commercialization. This highlights XAI's utility as a tool for enhancing each creative phase of innovation.

Second, the study highlights the importance of collaboration between technical teams and domain experts. Innovation managers should establish interdisciplinary teams composed of both technical and domain experts as XAI's explainability bridges communication gaps, ensuring that insights generated by the system are actionable for non-technical staff. In this way, a virtuous feedback cycle can be generated between domain experts and XAI systems, so that insights gathered from in-real-life testing of XAI-driven ideas can help refine algorithms, ensuring continuous improvement and alignment with innovation goals.

Third, the adoption of XAI during innovation changes resource allocation within organizations, hence requiring associated organizational change. Specifically, by sustaining knowledge-intensive tasks in all the phases of innovation, XAI allows organizations to redirect human and financial resources toward higher-value activities, such as refinement of innovative ideas and strategic decision-making. Accordingly, managers should consider restructuring teams to balance the roles of technical experts and domain specialists to better integrate XAI into innovation workflows. This reallocation also calls for new leadership strategies to manage interdisciplinary teams and foster a culture that embraces both technological and human contributions to innovation.

Fourth, the paper provides important practical implications on fostering human-machine collaboration, which could be useful for both managers and policymakers in designing initiatives to promote the so called *"hybrid intelligence"*. Hybrid intelligence is the combination of human and machine intelligences so that humans and machines collaborate to achieve better innovation performance than the isolated operation (Akata et al., 2020). We identified four main types of hybrid intelligence corresponding to the four creative phases of innovation. These hybrid intelligence types shape human-machine collaboration while mitigating automation bias and algorithm aversion, which are largely acknowledged as key obstacles to effective human-AI collaboration. In the *preparation* phase, we found that XAI provides *"autonomous intelligence"*. Specifically, the XAI system adapts to different situations and acts autonomously without human assistance to acquire, process, and organize vast amounts of data from diverse sources. Autonomous intelligence minimizes human effort in early-stage data handling and allows teams to focus on the subsequent phases of innovation. For policymakers, the integration of autonomous intelligence in early-stage innovation processes highlights the need for regulatory frameworks that ensure data quality and security. Policies should encourage transparency in how XAI systems collect and process information. In the *incubation* phase, we found that XAI provides *"automated intelligence"*, consisting in the automation of the generation and evaluation of potential ideas. By analyzing existing data and producing actionable insights without human intervention, automated intelligence accelerates the cycle of idea refinement for managers and ensures that the most promising concepts are ready for further exploration. At the same time, policymakers should establish standards for transparency in XAI-driven idea generation and evaluation, ensuring that automated intelligence does not reinforce existing biases or exclude unconventional but high-potential ideas. Moreover, policymakers need to address legal and ethical concerns around ownership and accountability of XAI-generated ideas, ensuring fair distribution of rights between AI and organizations. In the *illumination* phase, we found that XAI provides *"augmented intelligence"*, as the XAI system extends human capabilities and continuously learns from their interactions with humans and the environment. Indeed, in the illumination phase XAI works alongside humans to test ideas and apply knowledge in real-world settings. In this process, explainability supports managers in interpreting and testing results, while interactive feedback loops help refine both the ideas and the underlying algorithms. Regulations should ensure that XAI explanations are clear and accessible to all users, preventing misinterpretation of XAI-generated insights, especially in critical industries, such as healthcare or finance. Finally, in the *verification* phase, we found that XAI provides *"assisted intelligence"*, where XAI assists humans in monitoring KPIs, assessing market readiness, and suggesting actionable adjustments through specific systems. While humans retain control, XAI's assistance ensures that the transition from innovation to implementation is efficient and informed, enabling organizations to achieve their strategic goals effectively. Since assisted intelligence supports but does not replace human judgment, policymakers should promote guidelines that prevent organizations from blindly implementing AI-generated insights. Requiring human oversight and documented justification for AI-assisted decisions in regulatory filings could help balance AI efficiency with responsible governance.

5.3. Limitations and future research

While this study contributes to understanding how XAI influences knowledge management along the four creative phases of innovation, it is not without limitations.

First, the study is based on a single case study. Although this approach provides valuable insights for theory building (Yin, 2014), empirical validation is necessary to establish the model's robustness and applicability across diverse organizational contexts. Future research could employ surveys or experiments to generalize and test the proposed framework.

Second, this study primarily focuses on the benefits of XAI within the innovation process, with limited attention to the potential challenges and trade-offs in this integration. For instance, the implementation of XAI systems may lead to organizational resistance, ethical concerns, or unintended biases in decision-making. Future research could explore these barriers and develop strategies for mitigating their impact.

Third, the projects analyzed in the paper are conducted by R&D-intensive firms that highly revolve around product innovation. The results of the study may be different when considering innovation in service firms, such as healthcare or finance. Future work could examine how the integration of XAI aligns with the unique demands and constraints of the service industry.

Finally, the ethical and social implications of incorporating XAI into innovation processes warrant further investigation. Questions related to data privacy, transparency, and the potential for XAI to replace human roles in certain aspects of innovation need to be critically examined. Future studies could investigate frameworks for ethical XAI implementation that balance efficiency and innovation with accountability and inclusivity.

CRediT authorship contribution statement

Ilaria Mancuso: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Antonio Messeni Petruzzelli:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Umberto Panniello:** Writing – review & editing, Validation, Supervision, Methodology, Data curation, Conceptualization. **Federico Frattini:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

Nothing to declare.

Appendix A. Interview Protocol.

1. Describing Intellico's Business Model
   - What are Intellico's key resources and competencies?
   - What are Intellico's key activities and processes?
   - Who are Intellico's customers (e.g., incumbents, SMEs)?
   - What products/services does Intellico offer to its customers (i.e., offerings)? Could you focus on the custom solutions Intellico provides to its clients and those utilizing XAI, in terms of employed technologies and specific functionalities?
2. Specific Case Study: Development of XAI Applications (for custom projects)
   - Can you provide a general profile of the Intellico client company that adopted its XAI solution(s)? Why did the client company approach Intellico, i.e., what problem did it want to solve, or what process did it aim to optimize?
   - Could you describe the process of adopting the XAI solution for the client? What business processes were improved using Intellico's solutions?
   - What are the initial steps you take when starting a new XAI application development project with a client?
   - How is client knowledge gathered during this phase (e.g., workshops, interviews, brainstorming sessions)?
   - What types of data and inputs are required from the client for the development of the XAI application? How are these data collected and managed?
   - How is the client involved in the development and testing process of the XAI application? How is the human-technology interaction tested during the validation phase?
   - What feedback is requested from the client, and how are project revisions handled? How is client knowledge incorporated into the iterations of the XAI application?
   - How is the application adapted to changes in the client's processes?
3. Specific Case Study: XAI Application and Knowledge Management
   - How has Intellico's XAI solution facilitated knowledge management within the client company? How has Intellico's XAI-based solution contributed to collecting, organizing, and sharing knowledge?
   - Which subprocesses in knowledge management were modified due to Intellico's XAI solutions (e.g., problem-solving, idea generation)? Specifically, can you detail how the subprocess worked before and after implementing the XAI solution and what changes were introduced by the XAI solution?
   - How were the XAI solutions integrated into existing business processes to improve knowledge management?
   - What specific XAI techniques are used to make AI models transparent? How is it decided which explanations to provide to users?
   - How do the explanations provided by XAI improve the client's understanding and use of data (e.g., enhancing business decisions)?
   - How are AI-generated explanations presented to users (e.g., visualizations, reports, or other tools)? What methods are employed to ensure the information provided by XAI is understandable and useable for the users?
4. Specific Case Study: Human-XAI Integration
   - What practices are adopted to ensure a smooth and satisfying human-technology interaction (e.g., usability testing, monitoring user experience)?
   - What tools or interfaces are made available to allow users to interact with XAI solutions (e.g., dashboards, APIs, user interfaces)?
   - How has user feedback been managed to improve the accuracy and explainability of XAI?
   - How was it ensured that the explanations provided are intuitive for users with varying levels of technical expertise? How do you guarantee these explanations are easily understandable?
5. Specific Case Study: Challenges and Opportunities
   - What were the main challenges Intellico faced in implementing XAI solutions for the client? What strategies were adopted to overcome these challenges?
   - Is there anything else you would like to add or discuss that we have not covered and you believe is relevant?

Data availability

No data was used for the research described in the article.

References

Akata, Z., Balliet, D., De Rijke, M., Dignum, F., Dignum, V., Eiben, G., … Welling, M. (2020). A research agenda for hybrid intelligence: Augmenting human intellect with collaborative, adaptive, responsible, and explainable artificial intelligence. *Computer, 53*(8), 18–28.

Al-Emran, M., Mezhuyev, V., Kamaludin, A., & Shaalan, K. (2018). The impact of knowledge management processes on information systems: A systematic review. *International Journal of Information Management, 43*, 173–187.

Alavi, M., & Leidner, D. E. (2001). Knowledge management and knowledge management systems: Conceptual foundations and research issues. *MIS Quarterly*, 107–136.

Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J. M., Confalonieri, R., … Herrera, F. (2023). Explainable artificial intelligence (XAI): What we know and what is left to attain trustworthy artificial intelligence. *Information Fusion, 99*, Article 101805.

Allen, R., & Choudhury, P. (2022). Algorithm-augmented work and domain experience: The countervailing forces of ability and aversion. *Organization Science, 33*(1), 149–169.

Amabile, T. M., Conti, R., Coon, H., Lazenby, J., & Herron, M. (1996). Assessing the work environment for creativity. *Academy of Management Journal, 39*(5), 1154–1184.

Andriani, P., Ali, A., & Mastrogiorgio, M. (2017). Measuring adaptation and its impact on innovation, search, and problem solving. *Organization Science, 28*(2), 320–338.

Aydin, H. (2021). Market orientation and product innovation: The mediating role of technological capability. *European Journal of Innovation Management, 24*(4), 1233–1267.

Botega, L. F. D. C., & da Silva, J. C. (2020). An artificial intelligence approach to support knowledge management on the selection of creativity and innovation techniques. *Journal of Knowledge Management, 24*(5), 1107–1130.

Brem, A., Giones, F., & Werle, M. (2021). The AI digital revolution in innovation: A conceptual framework of artificial intelligence technologies for the management of innovation. *IEEE Transactions on Engineering Management, 70*(2), 770–776.

Del Giudice, M., Scuotto, V., & Papa, A. (2023). *Knowledge management and AI in society 5.0*. Routledge.

Demarest, M. (1997). Understanding knowledge management. *Long Range Planning, 30*(3), 374–384.

Dileo, M., Olmeda, R., Pindaro, M., & Zignani, M. (2024). Graph machine learning for fast product development from formulation trials. In *Joint European conference on machine learning and knowledge discovery in databases* (pp. 303–318). Cham: Springer Nature Switzerland.

Elsbach, K. D., & Stigliani, I. (2019). New information technology and implicit bias. *Academy of Management Perspectives, 33*(2), 185–206.

European Commission. (2017). *Guidelines on automated individual decision-making and profiling for the purposes of regulation 2016/679*.
Fowler, A. (2000). The role of AI-based technology in support of the knowledge management value activity cycle. *The Journal of Strategic Information Systems, 9*(2–3), 107–128.
Gama, F., & Magistretti, S. (2025). Artificial intelligence in innovation management: A review of innovation capabilities and a taxonomy of AI applications. *Journal of Product Innovation Management, 42*(1), 76–111.
Garbuio, M., & Lin, N. (2021). Innovative idea generation in problem finding: Abductive reasoning, cognitive impediments, and the promise of artificial intelligence. *Journal of Product Innovation Management, 38*(6), 701–725.
Ghasemaghaei, M., & Calic, G. (2020). Assessing the impact of big data on firm innovation performance: Big data is not always better data. *Journal of business research, 108*, 147–162.
Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. *Organizational Research Methods, 16*(1), 15–31.
Hamm, P., Klesel, M., Coberger, P., & Wittmann, H. F. (2023). Explanation matters: An experimental study on explainable AI. *Electronic Markets, 33*(1), 17.
Haque, A. B., Islam, A. N., & Mikalef, P. (2023). Explainable artificial intelligence (XAI) from a user perspective: A synthesis of prior literature and problematizing avenues for future research. *Technological Forecasting and Social Change, 186*, Article 122120.
Jarrahi, M. H., Askay, D., Eshraghi, A., & Smith, P. (2023). Artificial intelligence and knowledge management: A partnership between human and AI. *Business Horizons, 66*(1), 87–99.
Kannelønning, M. S. (2024). Navigating uncertainties of introducing artificial intelligence (AI) in healthcare: The role of a Norwegian network of professionals. *Technology in Society, 76*, Article 102432.
Kim, D., Song, Y., Kim, S., Lee, S., Wu, Y., Shin, J., & Lee, D. (2023). How should the results of artificial intelligence be explained to users?-Research on consumer preferences in user-centered explainable artificial intelligence. *Technological Forecasting and Social Change, 188*, Article 122343.
Kolbjørnsrud, V. (2024). Designing the intelligent organization: Six principles for human-AI collaboration. *California Management Review, 66*(2), 44–64.
Konar, A. (2018). *Artificial intelligence and soft computing: Behavioral and cognitive modeling of the human brain*. CRC press.
Kulkov, I. (2021). The role of artificial intelligence in business transformation: A case of pharmaceutical companies. *Technology in Society, 66*, Article 101629.
Lane, J. N., Ganguli, I., Gaule, P., Guinan, E., & Lakhani, K. R. (2021). Engineering serendipity: When does knowledge sharing lead to knowledge production? *Strategic Management Journal, 42*(6), 1215–1244.
Leoni, L., Gueli, G., Ardolino, M., Panizzon, M., & Gupta, S. (2024). AI-Empowered KM processes for decision-making: Empirical evidence from worldwide organisations. *Journal of Knowledge Management, 28*(11), 320–347.
Mancuso, I., Petruzzelli, A. M., & Panniello, U. (2024). Value creation in data-centric B2B platforms: A model based on multiple case studies. *Industrial Marketing Management, 119*, 1–14.
Mancuso, I., Petruzzelli, A. M., Panniello, U., Frattini, F., & Del Giudice, M. (2024). Nurturing strategic agility through corporate venturing advisory: An exploratory analysis. *Industrial Marketing Management, 121*, 1–15.
Mancuso, I., Petruzzelli, A. M., Panniello, U., & Nespoli, C. (2023). A microfoundation perspective on business model innovation: The cases of roblox and meta in metaverse. *IEEE Transactions on Engineering Management, 71*, 12750–12763.
Mersha, M., Lam, K., Wood, J., AlShami, A., & Kalita, J. (2024). Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction. *Neurocomputing*, Article 128111.
Meske, C., Bunde, E., Schneider, J., & Gersch, M. (2022). Explainable artificial intelligence: Objectives, stakeholders, and future research opportunities. *Information Systems Management, 39*(1), 53–63.
Minh, D., Wang, H. X., Li, Y. F., & Nguyen, T. N. (2022). Explainable artificial intelligence: A comprehensive review. *Artificial Intelligence Review*, 1–66.
Misseyanni, A., Lytras, M. D., Papadopoulou, P., & Marouli, C. (Eds.). (2018). *Active learning strategies in higher education: Teaching for leadership, innovation, and creativity*. Emerald Publishing Limited.
Pietronudo, M. C., Croidieu, G., & Schiavone, F. (2022). A solution looking for problems? A systematic literature review of the rationalizing influence of artificial intelligence on decision-making in innovation management. *Technological Forecasting and Social Change, 182*, Article 121828.
Pinho, I., Rego, A., & Pina e Cunha, M. (2012). Improving knowledge management processes: A hybrid positive approach. *Journal of Knowledge Management, 16*(2), 215–242.
Raghu, T. S., & Vinze, A. (2007). A business process context for Knowledge Management. *Decision Support Systems, 43*(3), 1062–1079.
Sadeghi, K., Ojha, D., Kaur, P., Mahto, R. V., & Dhir, A. (2024). Explainable artificial intelligence and agile decision-making in supply chain cyber resilience. *Decision Support Systems, 180*, Article 114194.
Savino, T., Messeni Petruzzelli, A., & Albino, V. (2017). Search and recombination process to innovate: A review of the empirical evidence and a research agenda. *International Journal of Management Reviews, 19*(1), 54–75.
Scholz, R. W., & Tietje, O. (2002). *Embedded case study methods: Integrating quantitative and qualitative knowledge*. Sage.
Sjödin, D., Frishammar, J., & Thorgren, S. (2019). How individuals engage in the absorption of new external knowledge: A process model of absorptive capacity. *Journal of Product Innovation Management, 36*(3), 356–380.
Stiennon, N., Ouyang, L., Wu, J., Ziegler, D., Lowe, R., Voss, C., … Christiano, P. F. (2020). Learning to summarize with human feedback. *Advances in Neural Information Processing Systems, 33*, 3008–3021.
Tchuente, D., Lonlac, J., & Kamsu-Foguem, B. (2024). A methodological and theoretical framework for implementing explainable artificial intelligence (XAI) in business applications. *Computers in Industry, 155*, Article 104044.
Truong, Y., & Papagiannidis, S. (2022). Artificial intelligence as an enabler for innovation: A review and future research agenda. *Technological Forecasting and Social Change, 183*, Article 121852.
Van Lent, M., Fisher, W., & Mancuso, M. (2004). An explainable artificial intelligence system for small-unit tactical behavior. In *Proceedings of the national conference on artificial intelligence* (pp. 900–907). Menlo Park, CA: Cambridge, MA; London; AAAI Press; MIT Press, 1999.
Verganti, R., Vendraminelli, L., & Iansiti, M. (2020). Innovation and design in the age of artificial intelligence. *Journal of Product Innovation Management, 37*(3), 212–227.
Wang, P., & Ding, H. (2024). The rationality of explanation or human capacity? Understanding the impact of explainable artificial intelligence on human-AI trust and decision performance. *Information Processing & Management, 61*(4), Article 103732.
Wee, J. C. N., & Chua, A. Y. K. (2013). The peculiarities of knowledge management processes in SMEs: The case of Singapore. *Journal of Knowledge Management, 17*(6), 958–972.
Yin, R. K. (2014). *Case study research: Design and methods* (5th ed.). Thousand Oaks, CA: SAGE Publications.